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α-Keto Acid Dehydrogenase Complexes

XVII. Kinetic and Regulatory Properties of Pyruvate Dehydrogenase Kinase and Pyruvate Dehydrogenase Phosphatase from Bovine Kidney and Heart[1]

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Both pyruvate dehydrogenase (PDH) kinase and PDH phosphatase require a divalent cation for activity. Either Mg^{2+} or Mn^{2+}, but not Ca^{2+}, can satisfy this requirement. The apparent K_m of the kinase for $MgATP^{2-}$ is about 0.02 mM, whereas the apparent K_m of the phosphatase for Mg^{2+} is about 2 mM. The apparent K_m of the phosphatase for Mn^{2+} is about 0.5 mM. The bovine kidney and heart PDH kinases exhibit pH optima about 7.0–7.2 in the presence of either Mg^{2+} or Mn^{2+}. The pH optima of the corresponding PDH phosphatases are about 6.7–7.1 in the presence of Mg^{2+} and 7.5–7.6 in the presence of Mn^{2+}.

ADP is a competitive inhibitor of ATP. The apparent K_i for ADP is about 0.1 mM. No effect of adenosine 3′,5′-cyclic monophosphate (cyclic AMP) on either the kinase or the phosphatase was observed in these studies. Inhibition of PDH phosphatase by fluoride ion, inorganic orthophosphate, EDTA and EGTA is described.

Pyruvate protects the pyruvate dehydrogenase complex against inactivation by ATP, and this effect appears to be more pronounced with the bovine heart complex than with the bovine kidney complex. It appears that pyruvate exerts its inhibitory effect primarily on the PDH kinase. We have observed that the kidney kinase catalyzes a phosphorylation of casein. This reaction is insensitive to cyclic AMP, and it is inhibited by pyruvate and α-ketobutyrate, but not by α-ketoglutarate.

Dihydrolipoyl transacetylase markedly stimulates the rate of phosphorylation of PDH by PDH kinase. We have found that the transacetylase lowers the apparent K_m of the kinase for PDH from about 20 μM to about 0.6 μM.

The mammalian pyruvate dehydrogenase complex catalyzes a coordinated sequence of reactions (Reactions 1–4).

$$CH_3COCO_2H + \text{thiamine-PP} \longrightarrow CH_3CHOH\text{-thiamine-PP} + CO_2 \quad (1)$$

$$CH_3CHOH\text{-thiamine-PP} + [lipS_2]^5 \longrightarrow [CH_3CO\text{-S-lipSH}] + \text{thiamine-PP} \quad (2)$$

$$[CH_3CO\text{-S-lipSH}] + CoASH \longrightarrow CH_3CO\text{-S-CoA} + [lip(SH)_2] \quad (3)$$

$$[Lip(SH)_2] + DPN^+ \xrightarrow{(FAD)} [lipS_2] + DPNH + H^+ \quad (4)$$

$$\text{Sum: } CH_3COCO_2H + CoASH + DPN^+ \longrightarrow CH_3CO\text{-S-CoA} + CO_2 + DPNH + H^+ \quad (5)$$

Reactions 1 and 2 are catalyzed by pyruvate

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dehydrogenase (PDH),[5] Reaction 3 by dihydrolipoyl transacetylase, and Reaction 4 by dihydrolipoyl dehydrogenase (1, 2).

The activity of the complex is subject to regulation by phosphorylation and dephosphorylation of the pyruvate dehydrogenase component (3–8). Phosphorylation and concomitant inactivation of pyruvate dehydrogenase are catalyzed by a kinase, and dephosphorylation and concomitant reactivation are catalyzed by a phosphatase (3, 5, 7). The kinase, pyruvate dehydrogenase, and dihydrolipoyl dehydrogenase are tightly bound to the dihydrolipoyl transacetylase, whereas the phosphatase appears to be loosely associated with the complex (5, 7). This communication describes some of the kinetic and regulatory properties of PDH kinase and PDH phosphatase from bovine kidney and heart mitochondria.

MATERIALS AND METHODS

Materials

DPN, potassium pyruvate (Type III), imidazole (Grade III), 2-methylimidazole, 2-(*N*-morpholino)ethanesulfonic acid, *N*-tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid, and thiamine pyrophosphate were purchased from Sigma. Dithiothreitol was obtained from Calbiochem, adenine nucleotides and CoA came from P-L Biochemicals, and 2-(*N*-morpholino)propanesulfonic acid was from Schwarz-Mann. [γ-^{32}P]ATP was obtained from International Chemical and Nuclear Corp. All other chemicals were of the highest commercially available grades.

Highly purified preparations of the pyruvate dehydrogenase complex (sp act, 8–12 μmoles DPNH formed/min/mg protein at 30°C) from bovine kidney and heart mitochondria and component enzymes of these complexes were obtained as described previously (5). The kidney pyruvate dehydrogenase complex was generally stored at 4°C as a concentrated solution (30–50 mg/ml) in 0.05 M potassium phosphate buffer (pH 7.0) containing 0.1 mM EDTA and 2 mM dithiothreitol. The heart pyruvate dehydrogenase complex was kept at 4°C in a solution containing 0.01 M MOPS[5] buffer (pH 7.0), 0.016 M imidazole buffer (pH 7.0), 0.2 mM $MgCl_2$, and 1 mM dithiothreitol. For many experiments in which a different buffer system was desired, these solutions were diluted 10- to 40-fold with the desired buffer. For some experiments the original buffer was replaced by extensive dialysis against several changes of the desired buffer or by filtration through a column of Sephadex G-25 which had been equilibrated with the desired buffer.

The preparations of the kidney and heart pyruvate dehydrogenase complex contained tightly bound PDH kinase (4, 5). The PDH kinase is bound to the dihydrolipoyl transacetylase component of the complex. For certain experiments the complexes were resolved to obtain a transacetylase–kinase fraction or the free kinase (5).

Since the PDH phosphatase is loosely associated with the kidney and heart pyruvate dehydrogenase complexes (4, 5), purified preparations of the complexes contained variable amounts of the phosphatase. This "endogenous" PDH phosphatase was sufficient for many experiments. Otherwise, purified preparations of PDH phosphatase [sp act, 3500–9000 units/mg of protein (5)] were added. In general, the PDH phosphatase content of preparations of the pyruvate dehydrogenase complex could be reduced to a low level by ultracentrifugation at 144,000*g* for 2.5 hr to sediment the complex. Most of the PDH phosphatase remained in the supernatant fluid.

Assay of Pyruvate Dehydrogenase Complex (DPN-Reduction Assay)

The assay is based on spectrophotometric determination of the rate of formation of DPNH (Reaction 5). The standard assay mixture contained 50 μmoles of potassium phosphate buffer (pH 8.0), 0.2 μmole of thiamine pyrophosphate, 1.0 μmole of $MgCl_2$, 2.5 μmoles of DPN, 0.13 μmole of CoA, 2.6 μmoles of cysteine hydrochloride, 2.0 μmoles of potassium pyruvate, and enzyme complex in a total volume of 1.0 ml. The pH of the assay mixture was 7.5. The temperature was 30°C. The reaction was initiated by addition of enzyme complex. Activity is expressed as ΔA_{340}/min and is based on the initial rate.

Assay of PDH Kinase

The assay is based on the initial rate of inactivation of the pyruvate dehydrogenase complex in the presence of ATP and endogenous kinase (5). In the early stages of this investigation, the reaction mixture contained about 160 μg of pyruvate dehydrogenase complex, 0.02 M potassium phosphate

[5] Abbreviations used: PDH, pyruvate dehydrogenase component of the pyruvate dehydrogenase complex; [lipS_2], lipoyl moiety which is covalently bound to the transacetylase; EGTA, ethylene glycol bis(2-aminoethyl)-*N*,*N*′-tetraacetate; cyclic AMP, adenosine 3′,5′-cyclic monophosphate; cyclic GMP, guanosine 3′,5′-cyclic monophosphate; MOPS, 2-(*N*-morpholino)propanesulfonate; TES, *N*-tris(hydroxymethyl)-methyl-2-aminoethanesulfonate.

buffer (pH 7.5), 1 mM $MgCl_2$, 2 mM dithiothreitol, and 0.02–0.2 mM ATP, in a final volume of 0.2 ml. In later experiments, the reaction mixture was modified to contain 160 μg of enzyme complex, 0.02 M imidazole buffer (pH 7.0), 0.25–0.5 mM $MgCl_2$, 2 mM dithiothreitol, and 0.1 mM ATP. ATP was omitted from the control. The mixture was incubated at 30°C, and 0.02-ml aliquots were removed at 1-min intervals and diluted immediately into the DPN-reduction assay mixture. The initial rate of DPNH formation was determined spectrophotometrically. The difference (i.e., decrease) in A_{340}/min between the control and the experimental samples is a measure of kinase activity. Modifications of the kinase-assay components and conditions are noted in the text. In certain experiments PDH kinase activity was determined by measuring the initial rate of incorporation of ^{32}P-labeled phosphoryl groups from [γ-^{32}P]ATP into purified pyruvate dehydrogenase (5), the pyruvate dehydrogenase complex, or casein. Details of these experiments are given in the text. Aliquots (0.05 ml) were withdrawn from the incubation mixtures and were applied to squares (2 × 2 cm) of Whatman No. 3MM filter paper. The papers were washed four times with cold 10% trichloroacetic acid, twice with ethanol, and twice with ether (9). The papers were air dried and then counted in a liquid scintillation spectrometer.

Assay of PDH Phosphatase

The assay is a modification of that described previously (5). It is based on the initial rate of reactivation of inactivated (i.e., phosphorylated) preparations of the pyruvate dehydrogenase complex in the presence of the phosphatase and a divalent cation. In routine assays, 10 mM Mg^{2+} was used. Concentrated solutions of the pyruvate dehydrogenase complex were diluted to a protein concentration of about 1 mg/ml with 0.02 M imidazole buffer (pH 7.0), containing 0.25 mM $MgCl_2$ and 2 mM dithiothreitol. The diluted preparations were incubated at 30°C with 0.02 mM ATP (kidney) or 0.08–0.1 mM ATP (heart) until at least 95% of the DPN-reduction activity had disappeared. The incubation periods varied from 3 to 10 min for inactivation of the kidney pyruvate dehydrogenase complex and 20 to 60 min for the heart complex. Some preparations of the heart complex were partially inactivated in the absence of ATP during incubation at 30°C for longer than 30 min. In these cases, the incubation with ATP was allowed to proceed overnight at 4°C. The inactivated (i.e., phosphorylated) preparations of the pyruvate dehydrogenase complex were placed in an ice bath and were used within a 12-hr period. The phosphatase assay solution contained 80 μl of the inactivated complex, 10 μl of PDH phosphatase and 10 μl of 0.1 M $MgCl_2$. $MgCl_2$ or phosphatase, or both, were omitted from the controls. The mixture was incubated at 30°C for 2 min, and then a 20-μl aliquot was diluted immediately into the DPN-reduction assay mixture. The initial rate of DPNH formation at 30°C was determined spectrophotometrically. The difference (i.e., increase) in A_{340}/min between the control and the experimental samples is a measure of phosphatase activity. The extent of reactivation of the phosphorylated pyruvate dehydrogenase complex was limited to less than 50% to ensure linearity of the response (5). Modifications of the phosphatase assay components and conditions are noted in the text.

RESULTS

Effect of pH on PDH Kinase and PDH Phosphatase Activities

Initial rates of inactivation and reactivation of the pyruvate dehydrogenase complex in the presence of PDH kinase and PDH phosphatase, respectively, were measured as described in Methods. The bovine kidney and heart PDH kinases exhibited pH optima about 7.0–7.2 in the presence of either Mg^{2+} or Mn^{2+} (Figs. 1 and 2). In the presence of Mg^{2+}, the pH optimum of the kidney PDH phosphatase (6.7) appeared to be somewhat lower than that of the heart PDH phosphatase (7.1). The pH optimum of both phosphatases was shifted to about 7.5–7.6 in the presence of Mn^{2+}.

Effect of ATP Concentration on PDH Kinase Activity

The specificity of PDH kinase for ATP was reported previously (3). In the present investigation the apparent K_m value for ATP was determined to be about 0.02 mM for both the kidney and heart PDH kinases. The values ranged from 0.01 to 0.03 mM with different preparations of the kidney and heart pyruvate dehydrogenase complexes and with 0.02 M phosphate (pH 7.5), imidazole (pH 7.0), or MOPS (pH 7.0) buffers. Typical data are presented in Figs. 3 and 4 (Curve 1).

Inhibition of PDH Kinase by ADP

In a previous paper (4) evidence was presented that ADP protects the kidney and heart pyruvate dehydrogenase complexes

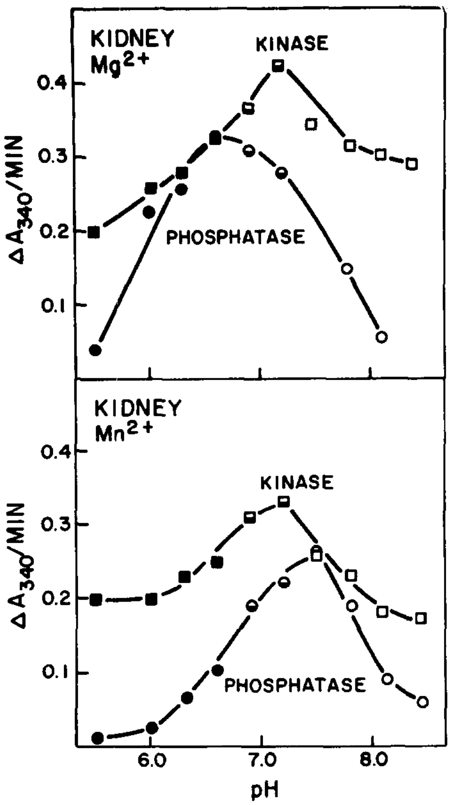

FIG. 1. Effect of pH on kidney PDH kinase and PDH phosphatase activities. Concentrated solutions of the active and inactive (i.e., phosphorylated) pyruvate dehydrogenase complex were diluted (40-fold) to a protein concentration of 1 mg/ml with 0.025 M imidazole–2-(N-morpholino)-ethanesulfonate buffers (■, ●) or 0.025 M imidazole-2-methylimidazole buffers (□, ○) at the desired pH. The PDH kinase and PDH phosphatase assays were carried out as described in Methods, with the following modifications. In the kinase assays, 0.25 mM $MgCl_2$ or 2.5 mM $MnSO_4$ was used, and the incubation period was 30 sec. For the phosphatase assays, 10 mM $MgCl_2$ or 2.5 mM $MnSO_4$ was used, and the incubation period was 1 min. In the phosphatase assays with Mn^{2+} as the divalent cation, 2.5 μg of kidney PDH phosphatase [sp act, 9000 units/mg (5)] was also added to each assay tube, since the preparation of the pyruvate dehydrogenase complex did not contain sufficient endogenous PDH phosphatase.

against inactivation by ATP. The data presented in Figs. 3 and 4 (Curves 1 and 2) show that ADP is competitive with ATP. The apparent K_i values for ADP varied from 0.1 to 0.3 mM with different preparations of the kidney and heart pyruvate dehydrogenase complexes. Similar results were obtained with phosphate, imidazole, and MOPS buffers.

AMP, GDP, malate, oxalacetate, succinate, citrate, isocitrate, lactate, alanine, phosphoenolpyruvate, L-carnitine, glutamate, and aspartate, at a concentration of 1 mM, had little effect, if any, on PDH kinase activity. Cyclic AMP (0.1–10 μM), cyclic GMP (0.1–10 μM), acetyl-CoA (0.5 mM), and acetoacetyl-CoA (0.1 mM) were also ineffective.

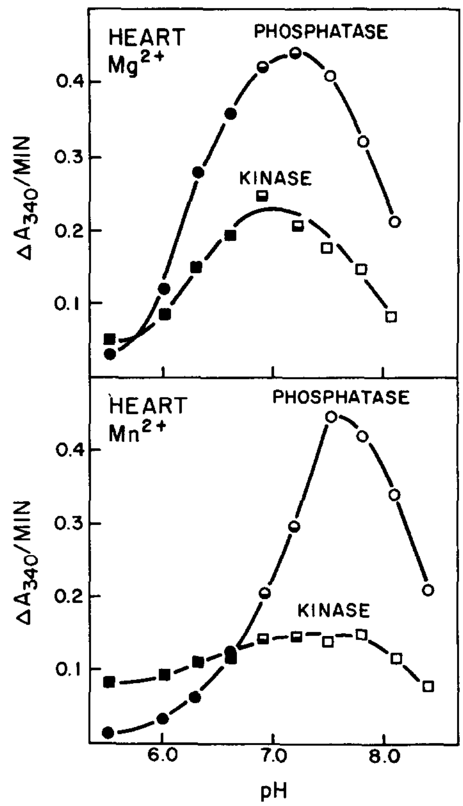

FIG. 2. Effect of pH on heart PDH kinase and PDH phosphatase activities. A preparation of the heart pyruvate dehydrogenase complex containing endogenous PDH phosphatase was used in these experiments. Conditions were as described in the legend of Fig. 1, except that 0.5 mM $MgCl_2$ was used in the PDH kinase assays, and the incubation period was 1 min.

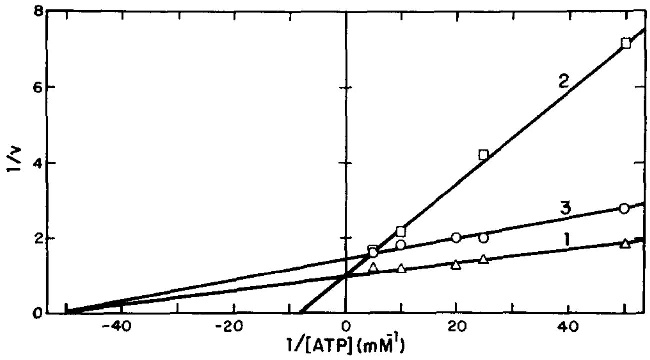

FIG. 3. Effects of ATP, ADP, and pyruvate on kidney PDH kinase activity. The reaction mixtures contained 150 μg of kidney pyruvate dehydrogenase complex, 4 μmoles of potassium phosphate buffer (pH 7.5), 0.2 μmole of $MgCl_2$, 0.4 μmole of dithiothreitol, and increasing amounts of ATP in a total volume of 0.2 ml (Curve 1). In parallel experiments, 0.1 μmole of ADP (Curve 2) and 0.1 μmole of potassium pyruvate (Curve 3) were included in the reaction mixtures. ATP was added last, and the mixtures were incubated at 30°C for 1 min. A 0.02-ml aliquot was removed and assayed for DPN-reduction activity.

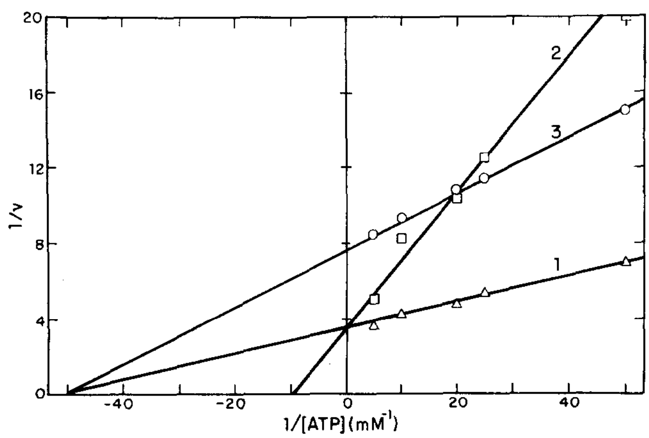

FIG. 4. Effects of ATP, ADP, and pyruvate on heart PDH kinase activity. Components and conditions were as described in Fig. 3, except that heart pyruvate dehydrogenase complex (150 μg) and 0.02 μmole of pyruvate were used, and the incubation period was 5 min.

Effect of Pyruvate on Inactivation of the Pyruvate Dehydrogenase Complex by ATP

Pyruvate protects the mammalian pyruvate dehydrogenase complex against inactivation by ATP (4). From the data presented in Figs. 3 and 4 (Curves 1 and 3) it appears that pyruvate is noncompetitive with ATP. The apparent K_i values for pyruvate varied from 0.9 to 2 mM with different preparations of the kidney pyruvate dehydrogenase complex and from 0.08 to 0.3 mM with different preparations of the heart complex. Further investigation revealed that pyruvate decreased the rate of incorporation of phosphoryl groups from [γ-^{32}P]ATP into the heart pyruvate dehydrogenase complex (Fig. 5), and thereby decreased the rate of inactivation of the complex.

The apparent difference in K_i values of the kidney and heart pyruvate dehydrogenase complexes for pyruvate suggested differences in the properties of the kidney and heart PDH kinases, the kidney and heart pyruvate dehydrogenases, or both sets of enzymes. Data reported previously (5, 6) indicate that the kidney and heart pyruvate dehydrogenases are very similar with respect

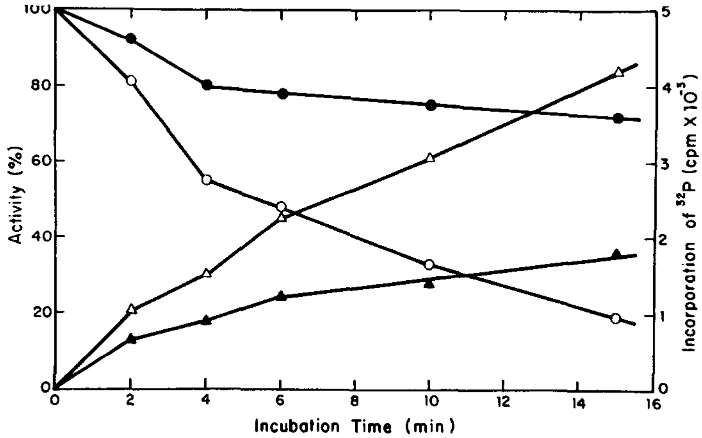

FIG. 5. Effect of pyruvate on the time course of phosphorylation of the heart pyruvate dehydrogenase complex. The incubation mixture contained 20 μmoles of phosphate buffer (pH 7.5), 1 μmole of $MgCl_2$, 2 μmoles of dithiothreitol, 0.05 μmole of [γ-^{32}P]ATP (about 5 mCi/mmole), and 0.34 mg of heart pyruvate dehydrogenase complex in a total volume of 1 ml. In a parallel experiment, 0.5 μmole of potassium pyruvate was included in the reaction mixture (●, ▲). ATP was added last, and the mixtures were incubated at 25°C. At the indicated times, 0.02-ml aliquots were assayed for DPN-reduction activity (○, ●) and 0.05-ml aliquots were assayed for protein-bound radioactivity (△, ▲).

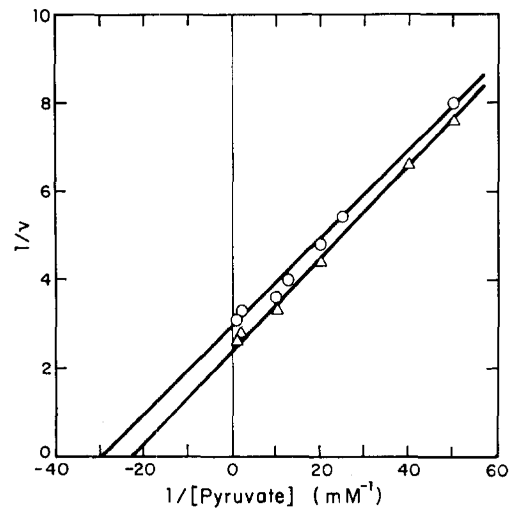

FIG. 6. Activity of heart (○) and kidney (△) pyruvate dehydrogenase complexes as a function of pyruvate concentration. The DPN-reduction assay was performed as described in Methods.

to molecular weight, subunit structure, and the appearance of peptide maps of tryptic digests. The apparent K_m values for pyruvate as substrate of the kidney and heart pyruvate dehydrogenases are also very similar, i.e., 0.044 and 0.035 mM, respectively (Fig. 6). Since the apparent K_i values for pyruvate differed by a factor of about 10, it appeared that the inhibitory effect of pyruvate on the phosphorylation of pyruvate dehydrogenase was exerted at a site other than the catalytic center of pyruvate dehydrogenase.

When kidney PDH kinase was added to an incubation mixture containing the heart pyruvate dehydrogenase complex and ATP, the response of the system to pyruvate resembled that observed with the kidney pyruvate dehydrogenase complex, i.e., the apparent K_i value for pyruvate was found to be about 1.1 mM. These observations suggested that the inhibitory effect of pyruvate was exerted on the PDH kinase rather than on the pyruvate dehydrogenase. Direct evidence for this conclusion was obtained from studies on the phosphorylation of casein by PDH kinase.

Phosphorylation of Casein by PDH Kinase

A search for alternative substrates for PDH kinase resulted in the finding that kidney PDH kinase catalyzes a phosphorylation of casein. The rate of this reaction is about 0.5% of that obtained with pyruvate dehydrogenase as substrate under similar conditions. Phosphorylation of casein by kidney

TABLE I

PHOSPHORYLATION OF CASEIN BY KIDNEY PDH KINASE[a]

Components	Incorporation of ^{32}P (cpm/0.05 ml)
Experiment 1	
Casein	20
Kinase (160 μg)	160
Kinase (320 μg)	185
Casein + kinase (160 μg)	700
Casein + kinase (320 μg)	1380
Kinase (160 μg) + pyruvate	70
Casein + kinase + pyruvate	100
Experiment 2	
Casein	0
Kinase (150 μg)	375
Kinase + casein	1450
Kinase + casein + cyclic AMP	1570

[a] The incubation mixtures contained 12.5 μmoles of potassium phosphate buffer (pH 7.5), 25 nmoles of [γ-^{32}P]ATP (about 10,000 cpm/nmole) and, where indicated, 0.5 mg of casein, 150–320 μg of kidney PDH kinase [sp act 320–385 units/mg (5)], 2.5 nmoles of cyclic AMP, and 0.25 μmole of potassium pyruvate in a total volume of 0.25 ml. The mixtures were incubated at 30°C for 30 min, and 0.05-ml aliquots were assayed for protein-bound radioactivity. The control contained radioactive ATP and phosphate buffer.

PDH kinase is insensitive to cyclic AMP, and it is inhibited by pyruvate (Table I). α-Ketobutyrate, which also serves as a substrate for the pyruvate dehydrogenase complex and protects the complex against inactivation by ATP (4), was about 50 % as effective (at 0.5 mM) as pyruvate, whereas α-ketoglutarate (at 0.5 mM) was ineffective. PDH kinase showed little activity, if any, toward calf thymus histone (Sigma Type II-A) in the presence or absence of cyclic AMP.

Effect of Dihydrolipoyl Transacetylase on PDH Kinase Activity

In a previous publication (5) we reported that the transacetylase markedly stimulated the rate of phosphorylation of pyruvate dehydrogenase by PDH kinase. We have found that the transacetylase lowers the apparent K_m of the kinase for pyruvate dehydrogenase about 30-fold (Fig. 7). The apparent K_m of the kidney PDH kinase for pyruvate dehydrogenase is about 20 μM. In the presence of the transacetylase, this value is decreased to about 0.6 μM. V appeared to be increased about 2-fold in the presence of the transacetylase. It should be noted that both the kinase and the pyruvate dehydrogenase are bound to the transacetylase in the pyruvate dehydrogenase complex.

Effect of Divalent Cations on PDH Kinase Activity

To remove Mg^{2+} which was used in the purification procedures (5), a solution (20–25

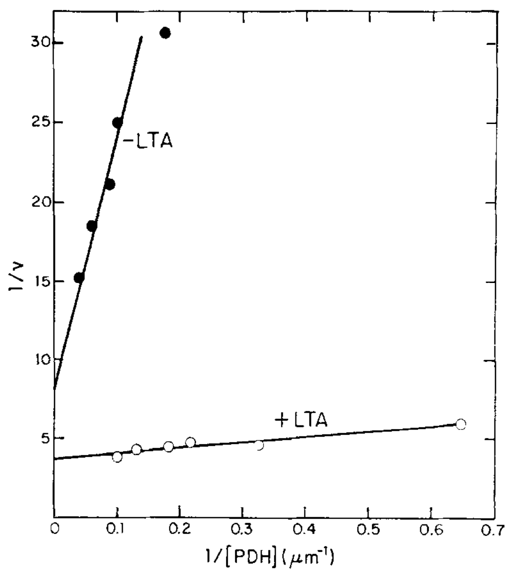

FIG. 7. Effect of dihydrolipoyl transacetylase on PDH kinase activity. The reaction mixtures contained 12.5 μmoles of potassium phosphate buffer (pH 7.5), 0.25 μmole of $MgCl_2$, 0.5 μmole of dithiothreitol, 0.025 μmole of [γ-^{32}P]ATP, and increasing amounts of pyruvate dehydrogenase (PDH) in a total volume of 0.25 ml. The molecular weight of PDH was taken to be 154,000 (6) In parallel experiments, 28 μg of kidney PDH kinase (●) and 7 μg of kidney kinase and 30 μg of kidney dihydrolipoyl transacetylase (LTA) (○) were included in the reaction mixtures. ATP was added last, and the mixtures were incubated at 30°C for 2 min. A 0.05-ml aliquot was removed and assayed for protein-bound radioactivity. Rates are expressed as nmoles of ^{32}P incorporated/2 min/0.25 ml. A larger amount of kinase (28 μg) was used in the absence of transacetylase in order to obtain measurable rates. These rates were divided by four to correspond to the rates obtained with the mixtures of kinase (7 μg) and transacetylase (30 μg).

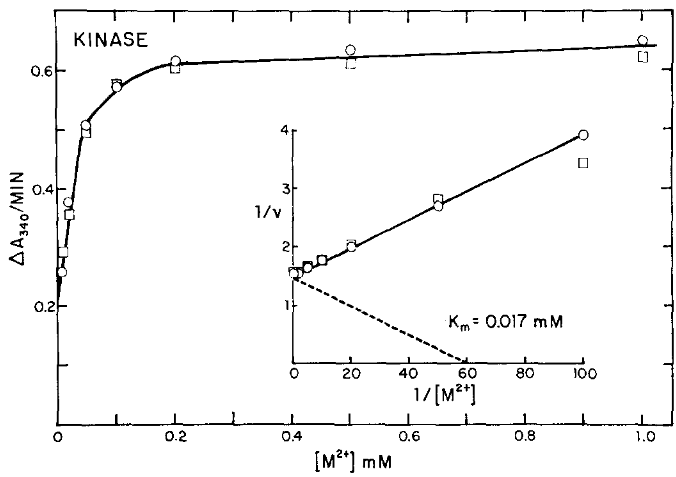

FIG. 8. Effect of Mg^{2+} and Mn^{2+} concentrations on kidney PDH kinase activity. A preparation of the kidney pyruvate dehydrogenase complex was treated with EDTA and passed through Sephadex G-25 as described in the text. The reaction mixtures contained 70 μg of enzyme complex, 0.02 μmole of ATP, 0.16 μmole of dithiothreitol, and increasing amounts of $MgCl_2$ (○) or $MnSO_4$ (□) in a total volume of 0.1 ml of 0.05 M TES buffer (pH 7.5). The mixtures were incubated at 30°C for 2 min prior to addition of ATP, and 30 sec after addition of ATP a 0.02-ml aliquot was removed and assayed for DPN-reduction activity.

mg of protein in 0.5 ml) of the kidney or heart pyruvate dehydrogenase complex was made 10 mM with respect to EDTA (pH 7.0), and after standing for 1 hr at 4°C the solution was passed through a column of Sephadex G-25 (1 × 10 cm) which had been equilibrated with 0.05 M TES[5] buffer (pH 7.5), containing 2 mM dithiothreitol. The response of the kidney PDH kinase to Mg^{2+} is shown in Fig. 8. The apparent K_m for Mg^{2+} (0.017 mM) was approximately equal to the apparent K_m for ATP (0.02 mM). This finding provides strong evidence that $MgATP^{2-}$ is the substrate for kidney PDH kinase. The results of other experiments with variable concentrations of Mg^{2+} and ATP were consistent with this conclusion and indicated that neither Mg^{2+} nor ATP exerted additional effects on the PDH kinase. Similar results were obtained with the heart PDH kinase. The kidney and heart PDH kinases were also active with Mn^{2+} as the divalent cation (Fig. 8). The apparent K_m for Mn^{2+} was about 0.017 mM. Ca^{2+} showed little ability, if any, to replace Mg^{2+} or Mn^{2+}. In the presence of 1 mM Mg^{2+}, 0.1–1.0 mM Ca^{2+} did not inhibit the kidney PDH kinase.

Effect of Divalent Cations on PDH Phosphatase Activity

With the phosphorylated (inactivated) pyruvate dehydrogenase complex from bovine kidney as the substrate and with 0.02 M imidazole buffer (pH 7.0), the apparent K_m of the kidney PDH phosphatase for Mg^{2+} varied from 1.8 to 3.4 mM. This variation may be due to the use of different preparations of the kidney complex and the kidney PDH phosphatase in these experiments, or possibly to the presence of small amounts of phosphate buffer (1–2 mM) in some preparations of the kidney complex. Typical data are presented in Fig. 9. With the heart PDH phosphatase acting on the kidney complex (phosphorylated), the apparent K_m for Mg^{2+} was about 1.7 mM. With the heart PDH phosphatase acting on the heart pyruvate dehydrogenase complex (phosphorylated), the apparent K_m for Mg^{2+} was about 2.0 mM (Fig. 10).

Both the kidney and heart PDH phosphatases were also active with Mn^{2+} as the divalent cation. The apparent K_m of the kidney PDH phosphatase for Mn^{2+} varied from 0.32 to 0.7 mM. Typical data are presented

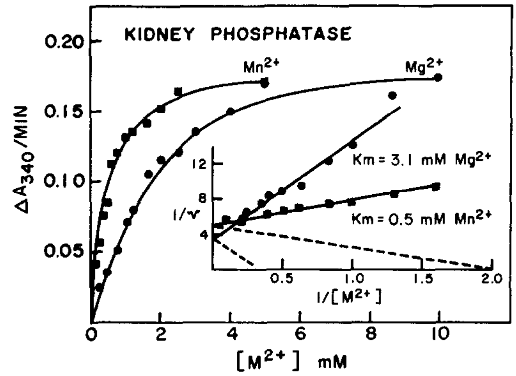

FIG. 9. Effect of Mg^{2+} and Mn^{2+} concentrations on kidney PDH phosphatase activity. Kidney PDH phosphatase [20 μg; sp act, 3500 units/mg (5)] and increasing amounts of either $MgCl_2$ (●) or $MnSO_4$ (■) were added to assay mixtures containing the inactivated (i.e., phosphorylated) pyruvate dehydrogenase complex from kidney in 0.02 M imidazole buffer (pH 7.0). PDH phosphatase assays were performed as described in Methods.

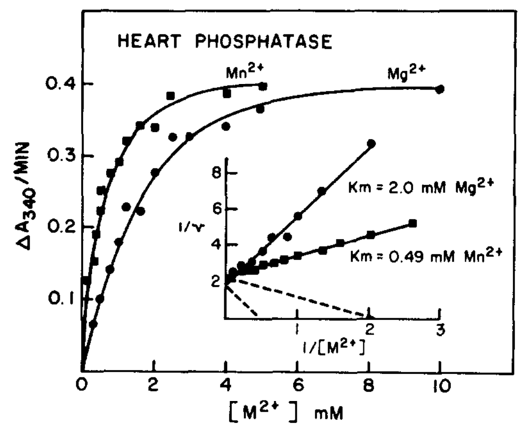

FIG. 10. Effect of Mg^{2+} and Mn^{2+} concentrations on heart PDH phosphatase activity. A preparation of the heart pyruvate dehydrogenase complex containing endogenous PDH phosphatase was used in these experiments. Conditions were as described in the legend of Fig. 9, except that PDH phosphatase was not added to the assay tubes.

in Fig. 9. With the heart PDH phosphatase, the apparent K_m for Mn^{2+} was about 0.5 mM (Fig. 10).

Other divalent metal ions were less effective than Mg^{2+} or Mn^{2+} (Fig. 11). At an optimum concentration of 0.1–0.2 mM, Ca^{2+} appeared to be about 10% and 25% as effective as Mg^{2+} or Mn^{2+} with the kidney and heart PDH phosphatases, respectively. Ca^{2+} appeared to inhibit the response of the kidney PDH phosphatase to Mg^{2+}. In the presence of 10 mM Mg^{2+}, 1 mM Ca^{2+} inhibited the kidney PDH phosphatase about 50%. At an optimum concentration of about 1 mM, Co^{2+} gave about 40–50% of the maximum response observed with Mg^{2+} or Mn^{2+} (Fig. 11). At concentrations above 1 mM, Co^{2+} was inhibitory. Zn^{2+} was inactive with the heart PDH phosphatase.

Inhibition of PDH Phosphatase

Fluoride ion markedly inhibited the bovine heart PDH phosphatase when Mg^{2+} (10 mM) was used as the divalent cation (Fig. 12). Fifty-per-cent inhibition was observed at a F^- concentration of about 0.8 mM. F^- was much less effective when Mn^{2+} was the divalent cation. Control experiments showed that NaF did not inhibit the activity of the pyruvate dehydrogenase complex in the DPN-reduction assay. Similar results were obtained in the present investigation with the bovine kidney PDH phosphatase and in an independent study by Wieland *et al.* (10) with the porcine heart PDH phosphatase. Since F^- has little effect, if any, on the PDH kinase (5), it can be used to selectively inhibit endogenous PDH phosphatase in preparations of the pyruvate dehydrogenase complex.

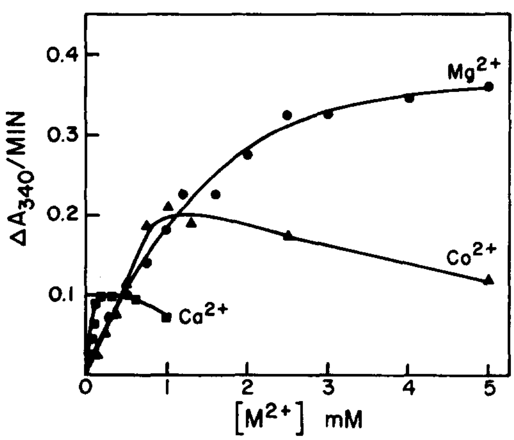

FIG. 11. Effect of Ca^{2+} and Co^{2+} on heart PDH phosphatase activity. Other components and conditions were as described in the legend of Fig. 10.

Inorganic orthophosphate inhibited the PDH phosphatase. The extent of inhibition was much greater with Mn^{2+} than with Mg^{2+} (Fig. 13). At a Mn^{2+} concentration of 5 mM, 50% inhibition of the heart PDH phosphatase was observed at a phosphate ion concentration of about 2.5 mM. Preliminary results indicated that manganous phosphate *per se* inhibited the phosphatase. In separate experiments it was shown that KCl, at concentrations up to 50 mM, had little effect, if any, on the activity of either PDH phosphatase or the pyruvate dehydrogenase complex.

In the presence of 10 mM Mg^{2+}, 1 mM EDTA inhibited the heart PDH phosphatase about 40%. With 0.5 mM EGTA[5] and 10 mM Mg^{2+}, the extent of inhibition was about 65% with the heart PDH phosphatase and about 82% with the kidney PDH phosphatase. In the presence of 5 mM Mn^{2+}, 0.5 mM EGTA inhibited the kidney and heart PDH phosphatases only slightly (<10%). Treatment of the kidney PDH phosphatase or the phosphorylated pyruvate dehydrogenase complex with 5 mM EGTA overnight at 4°C, followed by filtration on Sephadex G-25 did not affect the activity of the phosphatase in the presence of 10 mM Mg^{2+}.

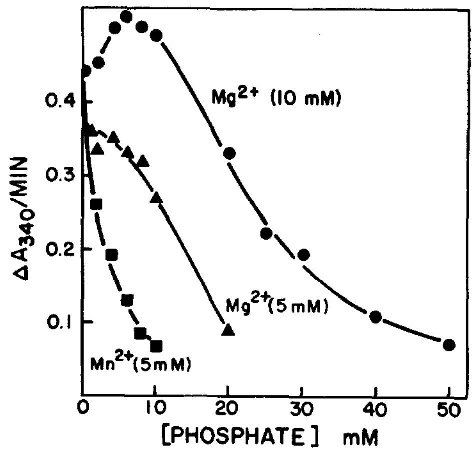

FIG. 13. Inhibition of heart PDH phosphatase by inorganic orthophosphate. A preparation of the heart pyruvate dehydrogenase complex containing endogenous phosphatase was inactivated (i.e., phosphorylated) as described in Methods. The PDH phosphatase assay mixtures contained, in addition to the inactivated complex (in 0.02 M imidazole buffer), either 5 mM $MgCl_2$ (▲), 10 mM $MgCl_2$ (●) or 5 mM $MnSO_4$ (■), and the indicated concentrations of potassium phosphate buffer (pH 7.0).

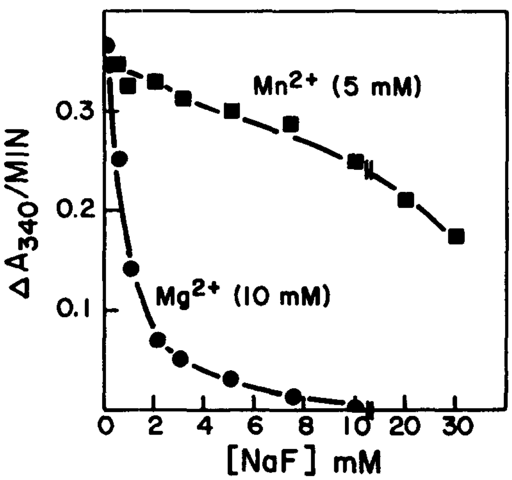

FIG. 12. Inhibition of heart PDH phosphatase by sodium fluoride. A preparation of the heart pyruvate dehydrogenase complex containing endogenous PDH phosphatase was inactivated (i.e., phosphorylated) and assays for PDH phosphatase were performed as described in Methods. The assay tubes contained, in addition to the inactivated complex [in 0.02 M imidazole buffer (pH 7.0)], either 10 mM $MgCl_2$ (●) or 5 mM $MnSO_4$ (■) and the indicated concentrations of NaF.

In the presence of 10 mM Mg^{2+}, acetyl-CoA (100 μM), acetoacetyl-CoA (50 μM), palmityl-CoA (50 μM), citrate and isocitrate (0.5 mM), and phosphoenolpyruvate (0.5 mM) had little effect, if any, on the activity of the bovine kidney or heart PDH phosphatase. Although cyclic AMP (1–100 μM) and cyclic GMP (1–50 μM) were tested under a variety of conditions, no effect of either of these compounds on the activity of the PDH phosphatase was observed. Preparations of PDH phosphatase at various stages of purification were used, including crude mitochondrial extracts, the preincubation time and the concentration of the phosphatase were varied, and ATP together with cyclic AMP was included in the preincubation mixture. When [γ-^{32}P]ATP was used, in the absence and presence of cyclic AMP, there was no detectable phosphorylation of the phosphatase preparations.

DISCUSSION

Both PDH kinase and PDH phosphatase require a divalent cation for activity. Either

Mg^{2+} or Mn^{2+}, but not Ca^{2+}, can satisfy this requirement. The apparent K_m of the kinase for $MgATP^{2-}$ is about 0.02 mM, whereas the apparent K_m of the phosphatase for Mg^{2+} is about 2 mM. This large difference could have physiological significance with respect to regulation of the activity of the phosphatase. The total content of magnesium in rat liver mitochondria is reported to be 20–25 nmoles/mg of protein (11). About one-half of this magnesium is in the matrix. This amount is approximately equal to the amount of mitochondrial adenine nucleotides (10–15 nmoles/mg protein) (12). The concentration of Mg^{2+} in the matrix may be determined, at least in part, by the mitochondrial ATP: ADP ratio (13), since ADP forms a much weaker complex with Mg^{2+} than does ATP.

ADP appears to be a rather weak competitive inhibitor of ATP. Since the apparent K_i of the kinase for ADP (about 0.1 mM) is larger than the apparent K_m of the kinase for ATP (about 0.02 mM), it does not appear that changes in the mitochondrial ATP: ADP ratio could be of physiological significance with respect to regulation of the activity of the kinase.

In our investigations, no effect of cyclic AMP was observed on the activity of either PDH kinase or PDH phosphatase, even though preparations of the two enzymes were tested at various stages of purification, including crude mitochondrial extracts. Wieland and Siess (7) also reported lack of an effect of cyclic AMP on the porcine heart PDH kinase. However, these investigators reported a cyclic AMP stimulation of the activity of the porcine heart PDH phosphatase, which they attributed to a hypothetical cyclic AMP-dependent PDH phosphatase kinase. Since there has been no confirmation of these results, in either our studies or those of other investigators (14, 15), and particularly since Siess and Wieland, in a recent publication (16), have qualified the interpretation of their previous observations, a possible role of cyclic AMP in regulation of the activity of PDH phosphatase appears doubtful.

No significant difference has been found thus far in the properties of the bovine kidney and heart PDH phosphatases. Both enzymes appear to consist of a single polypeptide chain of molecular weight about 100,000 (5). The phosphatase appears to be loosely associated with the bovine kidney and heart pyruvate dehydrogenase complexes. Inhibition of the bovine PDH phosphatase by fluoride ion, EDTA, and particularly EGTA, in the presence of excess Mg^{2+}, but not in the presence of excess Mn^{2+}, remains to be explained. A marked inhibition of the porcine heart PDH phosphatase by EGTA, but not by EDTA, in the presence of Mg^{2+}, and protection by Ca^{2+} against this inhibition, was reported by Wieland *et al.* (10). It is interesting to note that the affinity of EDTA for Mg^{2+} is about three orders of magnitude higher than is the affinity of EGTA for Mg^{2+} and that the affinity of both EDTA and EGTA for Mn^{2+} and Ca^{2+} is several orders of magnitude greater than is the affinity of EDTA for Mg^{2+} (17). Further understanding of these inhibitions must await analysis of the purified PDH phosphatase for a tightly bound metal ion.

PDH kinase, in contrast to PDH phosphatase, is tightly bound to the dihydrolipoyl transacetylase component of the bovine kidney and heart pyruvate dehydrogenase complexes (4, 5). This binding appears to markedly alter the kinetic properties of the kinase (T. E. Roche and L. J. Reed, unpublished observations). In the present investigation we have shown that the apparent K_m of the free kinase for its substrate, pyruvate dehydrogenase, is about 20 μM, whereas in the presence of the transacetylase the apparent K_m is decreased to about 0.6 μM.

Pyruvate protects the bovine kidney and heart pyruvate dehydrogenase complexes (4), and the porcine liver (4), heart, and brain complexes (8, 18) as well, against inactivation by ATP. That this inhibitory effect of pyruvate is exerted primarily, if not completely, on the PDH kinase is indicated by the observations that: (a) the kidney PDH kinase catalyzes a phosphorylation of casein and this reaction is inhibited by pyruvate; and (b) the apparent K_i values for pyruvate inhibition of the phosphorylation of kidney pyruvate dehydrogenase and casein by kidney transacetylase–kinase subcomplex are essentially identical (T. E. Roche and L. J. Reed, unpublished observations).

The inhibitory effect of pyruvate appears to be more pronounced with the bovine heart pyruvate dehydrogenase complex than with the kidney complex. Thus, apparent K_i values for pyruvate of 0.08–0.3 mM were found with the heart complex, and values of 0.9–2 mM were found with the kidney complex. The variability of these values suggests the possibility that the binding site for pyruvate on the PDH kinase is susceptible to modification during isolation of the complexes. The relatively low concentration of pyruvate required for significant inhibition of the bovine heart PDH kinase suggests that this mode of regulation may have physiological significance. It is interesting to note that the relative PDH kinase activity of the bovine heart pyruvate dehydrogenase complex is only about 25% of the kinase activity of the bovine kidney complex (5). Whether the heart complex contains less PDH kinase or a less active PDH kinase than does the kidney complex remains to be established. Attempts to purify the heart PDH kinase have met with only limited success thus far.

There does not appear to be a significant difference in the molecular activities of the kidney PDH phosphatase and PDH kinase. The molecular activities of both enzymes appear to be relatively low. Wieland *et al.* (10) have reported a molecular activity of about 4 moles of orthophosphate released from phosphorylated PDH per min per mole of porcine heart PDH phosphatase at 25°C. From a specific activity of about 100 nmoles of ^{32}P incorporated per min per mg of highly purified bovine kidney PDH kinase (5) and an assumed molecular weight of about 50,000, we estimate that the molecular activity of the kidney PDH kinase is about 5 moles of ^{32}P incorporated into PDH per min per mole of kinase at 30°C. In view of the low molecular activities of the kinase and the phosphatase, it seems that the resultant "ATPase activity" of this system is insignificant.

Possible regulatory effects that may result from the fact that the kinase is tightly bound to the transacetylase, whereas the phosphatase is loosely associated with the pyruvate dehydrogenase complex, and that both the kinase and the phosphatase act on a specific seryl residue in PDH (E. T. Hutcheson, J. R. Brown, and L. J. Reed, unpublished observations), remain to be evaluated.

Thus far we have not found a unique mechanism (e.g., interaction with cyclic AMP) for regulating the activity of the kinase or the phosphatase, nor have we found a specific metabolite which exerts reciprocal effects on the kinase and the phosphatase. If the concentration of free Mg^{2+} in the mitochondrial matrix actually changes during various metabolic states, the control exerted by this cation on PDH phosphatase activity could be physiologically significant. Regulation of the activity of PDH kinase, particularly the heart kinase, by pyruvate concentrations could also be significant. The activity of the pyruvate dehydrogenase complex is probably not regulated *in vivo* in an "on"–"off" manner by the kinase and the phosphatase. Rather, we visualize that these two antagonistic regulatory enzymes attain a steady state of activity which is dependent on the various factors mentioned above as well as factors yet to be determined.

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REFERENCES

1. Reed, L. J. (1969) *in* Current Topics in Cellular Regulation (Horecker, B. L., and Stadtman, E. R., eds.), Vol. 1, p. 233, Academic Press, New York.
2. Hayakawa, T., Kanzaki, T., Kitamura, T., Fukuyoshi, Y., Sakurai, Y., Koike, K., Suematsu, T., and Koike, M. (1969) *J. Biol. Chem.* **244**, 3660.
3. Linn, T. C., Pettit, F. H., and Reed, L. J. (1969) *Proc. Nat. Acad. Sci. U.S.A.* **62**, 234.
4. Linn, T. C., Pettit, F. H., Hucho, F., and Reed, L. J. (1969) *Proc. Nat. Acad. Sci. U.S.A.* **64**, 227.
5. Linn, T. C., Pelley, J. W., Pettit, F. H., Hucho, F., Randall, D. D., and Reed,

L. J. (1972) *Arch. Biochem. Biophys.* **148**, 327.

6. Barrera, C. R., Namihira, G., Hamilton, L., Munk, P., Eley, M. H., Linn, T. C., and Reed, L. J. (1972) *Arch. Biochem. Biophys.* **148**, 343.
7. Wieland, O., and Siess, E. (1970) *Proc. Nat. Acad. Sci. U.S.A.* **65**, 947.
8. Siess, E., Wittmann, J., and Wieland, O. (1971) *Hoppe-Seyler's Z. Physiol. Chem.* **352**, 447.
9. Mans, R. J., and Novelli, G. D. (1961) *Arch. Biochem. Biophys.* **94**, 48.
10. Wieland, O., Siess, E., v. Funcke, H. J., Patzelt, C., Schirmann, A., Löffler, G., and Weiss, L. *in* Metabolic Interconversion of Enzymes, Springer-Verlag, Berlin (in press).
11. Bogucka, K., and Wojtczak, L. (1971) *Biochem. Biophys. Res. Commun.* **44**, 1330.
12. Duée, E. D., and Vignais, P. V. (1969) *J. Biol. Chem.* **244**, 3920.
13. Blair, J. McD. (1970) *Eur. J. Biochem.* **13**, 384.
14. Jungas, R. L. (1971) *Metabolism* **20**, 43.
15. Coore, H. G., Denton, R. M., Martin, B. R., and Randle, P. J. (1971) *Biochem. J.* **125**, 115.
16. Siess, E. A., and Wieland, O. H. (1972) *Eur. J. Biochem.* **26**, 96.
17. Sillén, L. G., and Martell, A. E. (1971) Stability Constants of Metal-Ion Complexes, Supplement No. 1, The Chemical Society, London.
18. Wieland, O., and v. Jagow-Westermann, B. (1969) *FEBS Lett.* **3**, 271.